

Problem 2:

Create a dockerfile for nginx image with different html content and different nginx conf that listen to port 8080 instead of port 80 on the container

Create container from the new image

```
Dockerfile U x  index.html U  default.conf U
Dockerfile > ...
1 FROM nginx
2 COPY ./default.conf /etc/nginx/conf.d/default.conf
```

```
Dockerfile U  index.html U  default.conf U x
default.conf
1 server {
2     listen      8080;
3     listen  [::]:8080;
4     server_name localhost;
5
6     #access_log /var/log/nginx/host_access_log mai
```

```
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions$ docker build -t my_nginx .
[+] Building 1.5s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 98B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.14kB
=> [1/2] FROM docker.io/library/nginx:latest@sha256:0236ee02dcfce00b9bd83e0f5fbc51069e7e1161bd59d99
=> => resolve docker.io/library/nginx:latest@sha256:0236ee02dcfce00b9bd83e0f5fbc51069e7e1161bd59d99
=> [2/2] COPY ./default.conf /etc/nginx/conf.d/default.conf
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:08c07c01a94fec55a835f2d3b07e3c8aae6360e1736363628c85f7d76cd830b0
=> => exporting config sha256:6cd8bcflabe6b901c28557e09f9f156325ee975cc051ad67c304e160c81523f1
=> => exporting attestation manifest sha256:b531b9fc3232f8c2800c6af65cc851bfab13d0ed72f434512e86592
=> => exporting manifest list sha256:64f21a038c10fc4c9e243226e47b663c88f046e151a5a635eb32df0e6a9ca3
=> => naming to docker.io/library/my_nginx:latest
=> => unpacking to docker.io/library/my_nginx:latest
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions$ docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  E
centos:7                             be65f488b776                      301MB       76.1MB
image_name:latest                    72b1fe3a32cf                      237MB       62.9MB
my_nginx:latest                      64f21a038c10                      237MB       62.9MB
```

```

asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions$ docker run -d -p 8080:8080 my_nginx
15b19a9f9031a2c313384b96d79655edb2f9812a57e52d9800077bdda96d4c16
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CRE
PORTS         NAMES
15b19a9f9031   my_nginx   "/docker-entrypoint..." 14
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions$ docker exec -it 15b19a9f9031 bash
root@15b19a9f9031:/# curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>

```

Problem 3:

Create a reactjs simple app

Create a dockerfile to containerize the reactapp

Build the image and test it

(Bonus) create a dockerfile for the same app in smaller size using multi staging

```

Dockerfile > ...
1 FROM node:20-alpine
2 WORKDIR /app
3 COPY package.json .
4 RUN npm install
5 CMD [ "npm", "start" ]

```

```

.dockerignore
1 node_modules
2 dist
3 .git

```

```

asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_02/AngularPortfolio$ docker build -t myangularport .
[+] Building 120.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 128B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 62B
=> [1/4] FROM docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de0414
=> => resolve docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de0414

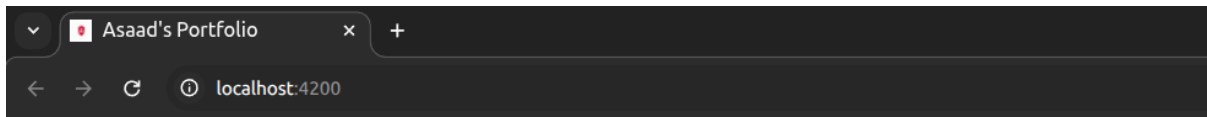
```

```

asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_02/AngularPortfolio$ docker run -v $(pwd):/app -p 4200:4200 myangularport:latest
> my-first-app@0.0.0 start
> ng serve

> Building...
✓ Building...
Browser bundles
Initial chunk files | Names      | Raw size
main.js             | main       | 27.33 kB |
styles.css          | styles     | 18.72 kB |

```



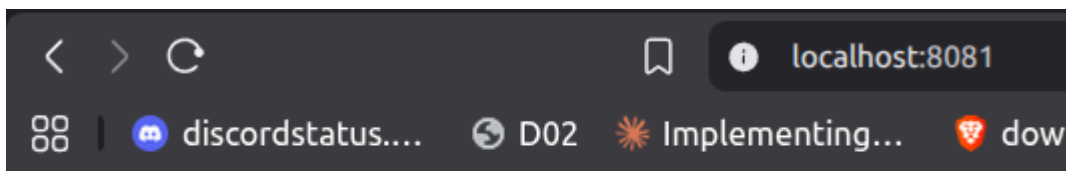
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Multi staging version

```
Dockerfile u x
Dockerfile > ...
1 FROM node:20-alpine AS builder
2 WORKDIR /app
3 COPY package.json .
4 RUN npm install
5 COPY . .
6 RUN npm run build
7
8 FROM nginx:alpine
9 COPY --from=builder /app/dist/myFirstApp/browser /usr/share/nginx/html

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS BRUNO LOGS POSTMAN CONSOLE
asaad@asus:~/ITI-OSAD/18 DOCKER/Solutions/Lab_02_02/AngularPortfolio (MultiStage)$ docker build -t myangularport-small .
[+] Building 9.3s (14/14) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default 0.0s
=> => transferring dockerfile: 237B                                              0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine                 0.6s
=> [internal] load metadata for docker.io/library/nginx:alpine                  0.6s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 62B                                                  0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142dde7315e3bc 0.1s
=> => resolve docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142dde7315e3bc0afa 0.1s
=> [internal] load build context                                                0.0s
=> => transferring context: 6.89kB                                              0.0s
=> CACHED [stage-1 1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557 0.1s
=> => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba 0.1s
```

```
/AngularPortfolio (MultiStage)$ docker run -p 8081:80 myangularport-small:latest
empty, will attempt to perform configuration
/docker-entrypoint.d/
.d/10-listen-on-ipv6-by-default.sh
checksum of /etc/nginx/conf.d/default.conf
en on IPv6 in /etc/nginx/conf.d/default.conf
```



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Problem 2:

Create a dockerfile for ubuntu image which sleeps by default for 5 sec or sleeps according to the given number in the docker command

```
Dockerfile U X
Dockerfile > ...
1 FROM ubuntu
2 ENTRYPOINT ["sleep"]
3 CMD ["5"]
```

Problem 3:

Push the images created in Problem 1&2 into your docker hub repo

```
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_02$ docker tag ubuntu:latest asaadmansour/ubuntu:latest
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_02$ docker push asaadmansour/ubuntu:latest
Using default tag: latest
The push refers to repository [docker.io/asaadmansour/ubuntu]
20043066d3d5: Mounted from library/ubuntu
561bf3bceeb5: Pushed
latest: digest: sha256:d0775219218d2038f0da606d72625952f4ccd55e805c28f1004c5e1e2ff898ab size: 855
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_02$
```

```
Login Succeeded
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker tag myangularport:latest asaadmansour/myangularport:latest
asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker push asaadmansour/myangularport:latest
Using default tag: latest
The push refers to repository [docker.io/asaadmansour/myangularport]
3e335cf2b31a: Pushed
589002ba0eae: Pushed
ed2fdcee5269: Pushed
9d10d4687fae: Pushed
729cb1ad6cc4: Pushed
aef53a46238b: Pushed
7de350c6a558: Pushed
7ad115895a6a: Pushed
latest: digest: sha256:5e8bd2a5cff3c530355c16cbb07f0969393752688b5f6e373f6f515c08f617c5 size: 856
```

```
asaadmansour/myangularport
asaadmansour/ubuntu:latest
```

Problem 4:

Create 2 nginx containers with network type bridge, enter to one of them and use curl command to view the content of the other container.

```
● asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker network create mynetwork
cfc040e23236c03ba4eceb9fba0f20d10b76247ccf994553d54d485fa99c9c69
● asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker run -d --network mynetwork nginx
819cd0b7b55992e4ef99accaa599ee10e96f910bd060937faf14a9958728d0a2
● asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker run -d --network mynetwork nginx
56d376b34ea7a72225d5d2e6034f38d1ce4ae5553e03c413ac84c92886e8c1f6
● asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker ps -n 3
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
56d376b34ea7   nginx         "/docker-entrypoint...." 31 seconds ago Up 31 seconds  80/tcp         nifty_khoran
819cd0b7b559   nginx         "/docker-entrypoint...." 39 seconds ago Up 38 seconds  80/tcp         determined_m
8400dad6dd3d   myangularport:latest "docker-entrypoint.s..." 29 minutes ago Exited (143) 27 minutes ago elegant_ritch
○ asaad@asus:~/ITI-OSAD/18_DOCKER/Solutions/Lab_02_03$ docker exec -it 56d376b34ea7 bash
root@56d376b34ea7:/# curl http://determined_mendeleev
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
```